

Same Text, Different Transitions: Action Insufficiency in Lean Provers

Anonymous ACL submission

Abstract

Neural Lean provers commonly condition tactic policies on a pretty-printed proof state, and some search procedures also use that rendering as node identity. We test the necessary assumption that the rendering is *action-sufficient*. Searches over Mathlib produce 70 repeat-verified alias classes and 103 paired state-tactic observations with identical text and opposite compiler labels, providing counterexamples to action sufficiency. Controlled rendering-preserving interventions identify notation-collapsed elaborated structure as a cause and reproduce the effect on a current Lean and Mathlib. A small learned acceptance model uses a compact structural channel to recover member-specific signal across unseen theorems within observed mechanisms. In a compiler-verified 96-pair stress benchmark, rendering-keyed retention discards a known available proof in nearly half of uniformly ordered alias arrivals, whereas a structural identity preserves every proof with 8% end-to-end overhead. Blind incidence and broad-search experiments separately measure how often evaluated policies encounter the demonstrated conditional failure.

1 Introduction

Neural Lean provers are given a textual rendering and asked to select a tactic. We show that this interface can map action-distinct states to the same text. Table 1 is one instance: changing only the order of `intro h` and `simp` reaches a common 206-byte rendering, after which another `simp` fails on one state and succeeds on the other. The outcomes repeat on fresh sessions.

Learned tactic generators for interactive theorem provers now close a substantial share of held-out library and competition problems (Polu and Sutskever, 2020; Han et al., 2022; Lample et al., 2022; Yang et al., 2023; Jiang et al., 2023), on benchmarks such as miniF2F (Zheng et al., 2022)

	history	then simp
s_1	<code>intro h; simp</code>	fails (×3)
s_2	<code>simp; intro h</code>	succeeds (×3)

Table 1: Two states in one Mathlib proof (`lintegral_ofReal_le_lintegral_nnnorm`), reached from a common ancestor by different tactic sequences. Both render to an identical 206-byte string; the same tactic, replayed three times per state on fresh sessions, deterministically fails on one and succeeds on the other.

and PutnamBench (Tsoukalas et al., 2024). Many share one interface decision: the policy conditions on a pretty-printed rendering of the proof state, and some best-first search implementations also use that text as the identity of a search node (Yang et al., 2023; Lample et al., 2022). These uses assume that the observation identifies the state’s response to an action, a transition-preservation condition required by safe state abstraction (Givan et al., 2003; Li et al., 2006). It is also implicit in learning tactic semantics from observed transitions (Samoylov and Vosoughi, 2025; Lamont et al., 2025). We test that assumption directly.

The issue also has a supervised-learning formulation. When the state rendering and tactic text are identical but compiler acceptance differs, a text-only acceptance predictor receives contradictory deterministic labels. The resulting error cannot be removed by model scale or optimization while the observation is fixed. This affects tactic-effect prediction, reranking, transition modeling, imitation learning, dataset construction, and evaluation wherever the rendered state is treated as a complete conditioning variable.

The hidden difference in Figure 1 is sixteen bytes of elaborated term: `ENNReal.ofNNReal x` versus `@WithTop.some.{0} NNReal x`. They are definitionally equal and both print as $\uparrow x$, but `simp`’s syntactic progress test distinguishes them.

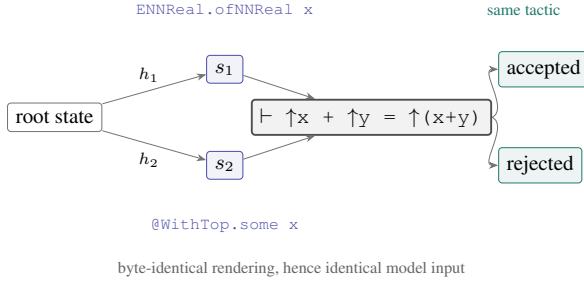


Figure 1: Observation aliasing. Two histories from one root reach states that differ only in elaborated structure the printer collapses; the rendering, and therefore the entire input supplied to the evaluated state-only generators, is byte-identical, yet the same tactic is accepted on one and rejected on the other.

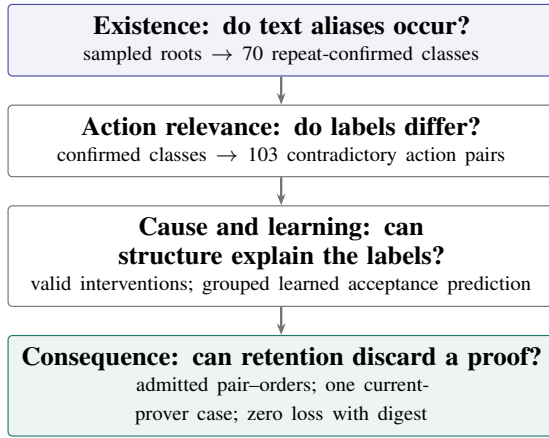


Figure 2: Conceptual evidence spine. Each arrow asks a separate question and uses the denominator printed beneath it. Population incidence, contradictory actions, structural learnability, and conditional search loss are estimated separately.

We call two such states *observation aliases*. Figure 2 organizes the evidence by logical implication and keeps incidence and consequence on their respective denominators. Its arrows ask distinct questions: whether aliases exist, whether the hidden difference changes an action, whether structural information explains and predicts the change, and whether an unsafe retention rule can discard a proof. Evidence for one arrow does not substitute for the denominator of another.

Over Mathlib (The mathlib Community, 2020) in Lean 4 (Moura and Ullrich, 2021) through Lean-Dojo (Yang et al., 2023), we make three contributions:

1. **Impossibility.** We give repeat-verified counterexamples to action sufficiency: 68 natural alias classes contain opposite acceptance labels for the same tactic, yielding 103 contra-

dictory textual inputs (§2–4). This conclusion is independent of the frequency with which a current policy encounters an alias.

2. **Empirical phenomenon.** Controlled interventions identify hidden elaborated structure as causal. A small acceptance model learns to use a compact structural channel, with theorem-grouped and mechanism-grouped evaluations separating interpolation from mechanism transfer (§5–7).

3. **Engineering consequence and repair.** A current step-prover and a 96-pair alias-conditioned stress benchmark demonstrate proof loss under unsafe retention. A structural identity key prevents every controlled loss at modest cost (§8).

Blind incidence, candidate exposure, and broad paired searches estimate the separate conditions under which the representational failure affects a search. Data, harnesses, and per-class outcomes are released.

2 Problem and scope

Let $T(s, a)$ denote the result of executing tactic a at Lean tactic state s , and Ψ the observable outcome (acceptance with a successor, rejection, or proof closure). Rejection is what the interaction API reports as an error, including Lean’s `made no progress`, the outcome most of the acceptance contradictions below take. A policy consumes $\varphi(s)$, the pretty-printed state; write $s_1 \sim_{\varphi} s_2$ when $\varphi(s_1) = \varphi(s_2)$.

The observation φ is *action-sufficient* if $s_1 \sim_{\varphi} s_2$ implies $\Psi(T(s_1, a)) = \Psi(T(s_2, a))$ for all a . This is a necessary one-step transition-consistency condition for abstractions such as bisimulation quotients and MDP homomorphisms (Givan et al., 2003; Li et al., 2006), which impose still stronger preservation requirements. We test only this one-step condition; its failure rules the stronger properties out. We distinguish acceptance, visible successor, and bounded-continuation equivalence throughout.

Acceptance ambiguity. For alias class C , some a has $r(s_i, a) \neq r(s_j, a)$, where $r \in \{0, 1\}$ is acceptance. The pair $(\varphi(s), a)$ then carries contradictory verified labels, and any acceptance predictor conditioned on it errs on at least $\min(n_0, n_1)/(n_0 +$

n_1) of members under the empirical member distribution. This lower bound applies to prediction of a specified action. The existence of a common successful tactic is a separate property.

Thus the corpus supplies compiler-verified contradictory supervision rather than annotation noise: Lean deterministically validates both labels on restored executable states. A text-only model can learn the average label distribution for an aliased input, while member-specific prediction requires an additional observation channel.

Candidate exposure and a direct model-facing test are reported in §7; the main text then evaluates search consequences.

Implementation scope. The observation claim is independent of a search implementation. The search-system claim is narrower: in the evaluated stack LeanDojo’s `TacticState` compares and hashes on the rendering with the executable handle excluded (`compare=False`), and the best-first prover keys nodes on that value, so $\sim_\varphi$ -equivalent nodes merge. LeanDojo-v2 uses Pantograph state handles and a history stack in its active prover, which avoids this particular merge. Its model prompt still uses the rendered state without the handle, so the action-sufficiency result continues to apply to policy conditioning. Appendix H gives the source and behavioral audits. A merge costs a proof when the discarded member’s bounded continuation language contains a proof absent from the retained member’s language.

3 Related work

Textual proof-state interfaces. The rendered-goal convention dates to CoqGym (Yang and Deng, 2019) and HOList (Bansal et al., 2019). miniCTX (Hu et al., 2024) and retrieval-augmented provers (Yang et al., 2023) show that context beyond the printed goal improves proving across problems. Our within-theorem members share the file, declaration, statement, and retrievable premises, so ordinary file context provides identical information for both members. LeanDojo (Yang et al., 2023), Pantograph (Aniva et al., 2025), and LeanTree (Kripner et al., 2025) define model and search interfaces without testing action sufficiency.

Structural and transition-grounded representations. Structural encodings are an established alternative to rendering, from graph representations of higher-order logic (Paliwal et al., 2020)

	blind	coerc.	struct.
roots replayed	1,025	1,423	3,882
expandable	644	947	2,553
with reconvergence	112	234	667
rendering-identical classes	150	321	893
flagged divergent	11	50	36
repeat-confirmed	3	47	21
new after deduplication	3	47	20
acceptance-ambiguous	3	46	19
contradictory pairs	6	78	19

Table 2: Discovery by sampling regime. Only the blind column estimates incidence; targeted columns provide examples for mechanism analysis. One cross-regime duplicate leaves 70 unique classes.

to Graph2Tac (Blaauwbroek et al., 2024) for Coq definitions. Samoylov and Vosoughi (2025) represent tactic semantics through induced state transformations, and 3D-Prover (Lamont et al., 2025) filters candidates by learned transition representations. Olejniczak et al. (2026) exploit proof-state symmetries. This work asks a logically prior question: whether the visible state is sufficient to define the transition that these methods model. Identical observations with contradictory deterministic transitions show that the answer can be negative. The same counterexample bounds text-based tactic-equivalence metrics such as LeanTutor’s (Patel et al., 2025).

Partial observability and search abstraction. Observation aliasing is the classical failure of a policy over observations to see its transition state (Whitehead and Ballard, 1991; Singh et al., 1994; Kaelbling et al., 1998). Tactic searches deduplicate states, the proof-search analogue of a transposition table (Zobrist, 1970); unsafe merging is the graph-history interaction problem (Kishimoto and Müller, 2004). Here elaboration creates the path dependence; game repetition rules are irrelevant.

4 Failure: search-reachable aliases

Method. From a root state we expand with a fixed tactic pool augmented by goal-derived rewrites and record, for every rendering, the set of *canonical* histories reaching it, where a step joins a history only if it changes the rendering (Appendix D). Classes reached by two prefix-free canonical histories are probed: one live state per history in a single session, the same battery for all members, and every flagged class re-verified on a fresh session with three repetitions per mem-

ber. States are read at five levels of printer explicitness by metaprogram probes through the ordinary interaction API (Appendix C); derived digests separate the elaborated goal (F_{core}), the environment (F_{env}), and executable identity. Configuration: Mathlib 29dcec07, Lean 4.10.0-rc1, LeanDojo 2.0.2. The structural column combines two targeted sweeps. The second selected 2,400 roots and restored 2,234; expansion counts exclude failed restorations.

Table 2 separates incidence from discovery. In the blind sample, 112/644 expandable roots reconverge to an identical rendering, but only 3 classes on 3/1,025 sampled roots are confirmed operationally divergent: 0.29% (95% CI 0.06–0.85%). The targeted regimes contribute 67 more unique classes and support mechanism analysis only. In total, the 70 classes span 53 theorems, 67 roots, and 40 files in nine top-level Mathlib directories. A corpus-wide cross-file census is dominated by declaration context and is reported separately (Appendix A).

Human scripts rarely revisit identical renderings; the aliases studied here arise primarily through search reconvergence (Appendix B).

Acceptance is contradictory. In 68 of 70 classes some tactic is accepted on one member and rejected on another, stably across three repetitions on fresh sessions; the remaining two diverge only in successor states. This yields 103 contradictory observation–action pairs: identical model input, identical tactic text, opposite verified acceptance. Any acceptance predictor conditioned on $(\varphi(s), a)$ errs on at least $\min(n_0, n_1)/(n_0 + n_1)$ of members per pair. Over the 103 pairs the member-weighted aggregate is 0.40, a conditional inconsistency floor restricted to these pairs. The phenomenon spans several elaboration mechanisms, whose distributions are reported below.

5 Cause: hidden elaborated structure

Reading each class at five levels of printer explicitness localizes the hidden content (Appendix C). The consumed rendering φ_{state} and its depth-expanded variant fail to separate any class, and the environment digest F_{env} also fails wherever the probe completes. This excludes ambient declaration context as the cause. Disabling notation under `pp.all` separates 67 of 70, showing that the notation layer collapses most of the hidden content (Ullrich and de Moura, 2020); depth elision has

no effect. The hidden content falls into six named categories plus two opaque classes: ENNReal coercion (56 classes), Fin eta-expansion (4), local-definition unfolding (4), instance paths (2), Ne versus Not (Eq ...) (1), and a proof-term difference (1). In the dominant family the separating rung differs by sixteen bytes, the same token pair every time. The divergence follows from tactic implementation. `simp`’s progress check compares the resulting term syntactically, so on the member holding the abbreviation it finds a rewrite and reports progress, while on the unfolded member it reports none and fails.

Rendering-preserving interventions. The classes above are observational. To manipulate the hidden structure directly we construct pairs of standalone examples (Figure 3) identical in every respect except a single `show` step replacing the goal with a definitionally equal term carrying different hidden syntax: same environment, binders, goal statement, and probe text. `show` accepts any term definitionally equal to the goal, so it alters the elaborated representation without altering what is proved. Validity is verified per pair in both directions: the intervention must have taken effect, so the variants’ structural digests must differ, and it must have remained hidden, so their pre-probe renderings must be byte-identical. Pairs failing either check are excluded and reported.

For the dominant coercion family 13 of 15 pairs pass both gates, and 11 of 13 change the probe outcome. The attempted Ne and eta interventions fail the byte-identity gate because their source spellings remain visible. The intervention therefore establishes a causal mechanism for the dominant family; the other observed families remain observational.

Visibility under continuation. In the original 59-class corpus, a shared continuation makes 18 classes visibly different, including 17 at `contrapose!`. A reduced battery that omits this operation yields near-total confinement. Escape is therefore a property of both the state and the tactic distribution (Appendix E).

Current stack. We repeat blind discovery on Lean v4.33.0-rc1 and contemporaneous Mathlib. Sixty uniformly sampled compiled files yield 2,507 states. From 600 fixed-seed roots, 570 restore successfully and 98 rendering-alias classes are found. Fresh repeated probing confirms two classes at one root, with seven con-

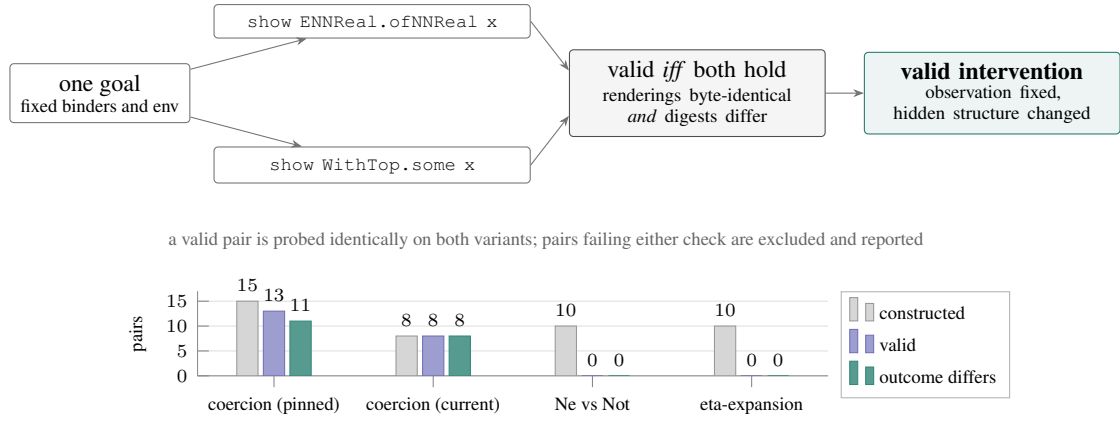


Figure 3: Rendering-preserving interventions. **Top:** the construction. A single `show` step replaces the goal with a definitionally equal term, so only the elaborated representation changes; a pair is used only if it passes both checks. **Bottom:** outcomes. The construction succeeds for the coercion family under the pinned stack and, with acceptance-sensitive probes, on a current Lean and Mathlib, where all 8 pairs are accepted on one variant and rejected on the other. The other two families yield zero valid pairs because the printer renders their two spellings differently.

tradictory class-action pairs. The resulting root incidence is 1/570, or .18% (Clopper–Pearson 95% CI .004–.97%). Both classes occur in `ModuleCat.disjoint_span_sum`; explicit printing separates terms produced by `rw` and `simp` only. In addition, all eight current coercion interventions pass both validity gates and reproduce opposite probe acceptance. Historical script replay reproduces one of eight older witnesses; the remaining scripts fail after API drift (Appendix G).

6 Learning from a structural channel

The observation aliases establish that the default rendering omits transition-relevant information. We next train a deliberately small model to predict tactic acceptance from a compact member-local structural channel. This test asks whether a learned predictor can use information unavailable in the rendering. Candidate exposure and zero-shot use by a frozen prover are evaluated separately in §7.

Representation. For each restored member we extract a sorted multiset of application-head constant paths from a shallow explicit rendering of the local context and goal. This *structural sketch* restricts its features to structural constants. Tactics, pair identities, histories, theorem names, and outcome labels are excluded. The sketch compresses the explicit rendering before it is given to the evaluator.

Learned acceptance model. The extractor succeeds for all 178 members of the 70 confirmed

classes. The acceptance evaluator selects one accepted and one rejected member for each repeat-confirmed class-action contradiction, giving 206 examples in 103 pairs from 51 theorems. We fit logistic regression over hashed sketch features and action-feature crosses (8,192 dimensions, $L_2=10^{-3}$, 500 gradient steps). The theorem-grouped protocol keeps every example from one theorem in a single fold. It estimates transfer to unseen theorems while allowing mechanisms to recur across folds. The mechanism-grouped protocol defines groups from the hidden elaborated delta: coercion construction, `Fin` eta-expansion, instance path, local-definition unfolding, and the residual printer-blind classes. Each group is held out in turn. The mechanism macro weights the four audited mechanisms equally; the all-group macro also includes the residual group. The `Ne-versus-Not` and proof-term classes yield different successful successors without opposite acceptance labels, so they fall outside this binary evaluation. The paired metric assigns 1 when the accepted member receives the higher score, 1/2 for a tie, and 0 for a reversal. Confidence intervals use the theorem as the resampling unit.

Within-mechanism generalization. Under theorem-grouped evaluation, the shallow sketch correctly orders 98 pairs, ties 5, and reverses none, giving a paired rank of .976 (theorem-cluster bootstrap 95% CI .943–1.000). It succeeds or ties on all 25 non-coercion pairs. The default rendering remains constant within every pair and therefore

Representation	classes split	pair rank	accuracy
default rendering	0/70	.500	.500
structural sketch	67/70	.976	.830
structural digest	70/70	n/a	n/a

Table 3: Theorem-grouped learned acceptance prediction on 103 contradictory pairs. Pair rank measures whether the accepting member is scored above the rejecting member. The digest is excluded from model features.

Held-out group	classes	pairs	pair rank
<code>Fin eta</code>	4	10	1.000
instance path	2	7	1.000
coercion	56	78	.718
local unfolding	4	4	.625
residual	2	4	.500
mechanism macro			.836
all-group macro			.769
pair weighted			.752

Table 4: Leave-one-group-out transfer in the primary 8,192-dimensional configuration. Each row trains on all other groups. The mechanism macro weights the four audited mechanisms equally; the all-group macro adds the residual group. The pair-weighted score retains the corpus distribution.

scores .500. The structural digest separates all classes and serves only as an opaque identity key.

Mechanism transfer. The theorem-grouped ranking ranges from .956 to .976 across hash dimensions from 2,048 to 16,384. Its primary estimate is pair weighted and the recurring coercion mechanism contributes 56 of 68 evaluated classes and 78 of 103 pairs. In the primary mechanism-held-out evaluation, the mechanism macro is .836, the all-group macro is .769, and the pair-weighted score is .752. The held-out `Fin` and instance-path groups each score 1.000, with four and two classes. Coercion scores .718, local unfolding .625, and the residual group .500. Feature hashing materially affects transfer: across four configurations, the mechanism macro ranges from .621 to .836, the all-group macro from .596 to .769, and the pair-weighted score from .199 to .850. The theorem-grouped result supports interpolation across unseen theorems within observed mechanisms. The mechanism-held-out scores provide preliminary evidence across audited mechanisms, while small groups and configuration sensitivity preclude a broad generalization to previously unseen alias forms.

Across all 178 members, the median sketch is

Input	all	coerc.	<code>Fin</code>	inst.	unfold	macro
ordinary rendering	.500	.500	.500	.500	.500	.500
+ structural sketch	.539	.564	.400	.429	.625	.504

Table 5: Frozen-prover paired ranking on 103 confirmed tactics. Each entry is the fraction for which the accepting member scores above the rejecting member, with ties assigned one half. The macro weights the four audited hidden-delta groups equally.

934 bytes (maximum 2,503), compared with 4,212 bytes (maximum 282,108) for shallow `pp.all`. Shallow sketches distinguish 67 of 70 classes and the digest distinguishes all 70. The learned model therefore establishes compact identifiability of the hidden signal within observed mechanisms. The prototype currently obtains an explicit rendering before compression, leaving direct extraction as an engineering problem.

7 Direct model-facing evaluation

The learned evaluator in §6 uses information in the sketch after task-specific training. We next test whether a current prover can use that channel without adaptation, using the same paired acceptance task.

Protocol. We freeze BFS-Prover-V2-7B (Xin et al., 2025) at one immutable snapshot. For each of the 103 repeat-confirmed class-action contradictions, the model scores the same tactic on one accepting and one rejecting member. The ordinary arm uses the model-card prompt `{state}:::`. The structural arm places the member-local sketch in a Lean comment between the rendering and the separator. We average the conditional log probabilities of the tactic tokens and count whether the accepting member receives the higher score; ties receive one half. The checkpoint, prompt, and scoring rule remain fixed. No outcome label, theorem, class, or mechanism from the alias corpus enters experiment-side training or calibration. We report every hidden-delta group separately and macro-average the four audited groups. Confidence intervals resample the 51 theorems.

Result. The structural input obtains .539 paired accuracy (theorem-cluster bootstrap 95% CI .440–.644), with 53 correct rankings, 5 ties, and 45 reversals. Its macro average across the four audited mechanisms is .504. The ten pairs with equal augmented-prompt token counts score .550. The ordinary arm ties exactly on every pair because the

k	mean $ A $	member cov.	ambiguous	headroom
1	1.0	.376	1	.000
2	2.0	.798	2	.000
4	4.0	.882	2	.000
8	7.5	.938	2	.000
16	11.9	.938	2	.000

Table 6: Candidate exposure over nested miniCTX beam prefixes. “Ambiguous” counts classes with an acceptance-divergent candidate. Headroom is the member-aware oracle coverage minus the best shared-action coverage. All 6,354 compiler calls agree within each state-candidate unit.

rendering and tactic are byte-identical. Appending the sketch to this frozen prover provides no reliable acceptance-sensitive signal. Together with the trained model’s .976 theorem-grouped score, this result decomposes structural-channel utility into information availability and model adaptation. The experiment conditions on a confirmed tactic; candidate generation and theorem-level proof rate remain outside its estimand.

Candidate exposure. One deterministic miniCTX (Hu et al., 2024) beam-16 list is generated for each of the 70 class renderings, with a 16-token tactic cap. Nested prefixes give $k = 1, 2, 4, 8, 16$, and every candidate is executed three times on all 178 members. Table 6 shows that wider prefixes improve general coverage through $k = 8$, while acceptance-divergent exposure remains confined to two classes. The member-aware and best shared-action oracles have identical coverage at every width, giving zero member-specific headroom. Thus the structural channel identifies member differences that the present candidate sets do not convert into a distinct available proof choice. Fixed beam-8 results for two additional generators and an auxiliary reranker are reported in Appendix K.

8 Consequence: search and retention

Aliasing matters to search only when a policy reaches a class and continuation quality differs across its members. We evaluate conditional loss after an alias encounter, natural continuation loss, and broad policy incidence separately.

Representative sensitivity. Every member in the original 59-class corpus is restored and searched independently under the same budget, model, and seed. With BFS-Prover-V2-7B (Xin et al., 2025), the first candidate list is byte-identical in all 59

Proposal generator	budgets	refined coverage	rendering loss
targeted	2/4/8/12	96/96 throughout	93/192 throughout
context bank	2/4/8/12	36/72/96/96	1/2, 1/2, 1/2, 93/192
generic first	2/4/8/12	96/96 throughout	60/192 throughout

Table 7: Exact uniform-order results on 96 admitted stress pairs. Loss is conditioned on the refined arm exposing a proof within budget. For the targeted generator, the pair-clustered 95% bootstrap interval is 46.4–50.0%; the eight-base-template interval is 46.9–49.5%.

classes; tactic execution alone then creates different trajectories. Only two classes yield distinct proof outcomes. In `snormEssSup_add_le`, the `intro h; simp member` is proved at expansion 14; the `simp; intro h member` remains unproved across repetitions and at triple budget. A rendering-keyed merge can therefore discard the member proved under this continuation. An ancestor-rooted control proves through another intermediate under either retention order, localizing the observed loss to the aliased node (Appendix J).

Alias-conditioned stress benchmark. We cross eight current-stack, rendering-preserving intervention pairs with 12 proposition contexts. Admission requires Lean to verify byte-identical default renderings, distinct structural digests, and a fixed continuation that closes one member and fails on the other. All 96 pairs pass. After both branches reach the alias, rendering identity retains the first arrival; rendering plus digest retains both. We enumerate both arrival orders exactly and also run 200 seeded random orders per pair under three proposal generators and four budgets.

The targeted generator exposes the known continuation and `rfl`. The refined arm proves all 96 pairs under both orders. Rendering identity loses the proof for 93 of 192 pair-order combinations, giving the exact uniform-order probability $93/(2 \cdot 96) = 48.4\%$; the randomized estimate at budget 2 is 48.0%. Table 7 shows that context-bank and generic-first generators change the conditional rate. This variation identifies proposal exposure as part of the estimand. The benchmark establishes a conditional failure of bounded-search completeness under controlled alias encounters. Its shared coercion mechanism prevents a population-level interpretation (Appendix J).

Broad nulls. Paired searches with rendering-only versus structurally refined identity prove the same theorem in all 158 runs (36 proved in both; one-sided 95% upper bound 1.9% for discordance).

Only 4 runs actually merge fingerprint-distinct states. Probing all 25 such merge pairs found no operational disagreement (upper bound 11.3%). Separately, replaying the known remainder of a Mathlib proof on 113 aliases loses none (upper bound 2.6%). These results constrain the broad rate under the tested policies and budgets. The 7B member search on the original corpus is outcome-informative in 2 of 59 classes and loses one local continuation. Policy exposure and candidate-set regret are reported in §7 and Appendix K.

Identity refinement. Model observation and node identity can differ. A 64-bit hash of the goal closed over its context and normalized for metavariables separates all 59 classes in the original corpus and prevents every stress-benchmark loss. In a rotated 24-theorem cost test, verbose printing raises end-to-end time by 179%, versus 8% for the digest (Appendices F and I). The digest is a version-specific, collision-prone prototype that provides no proof of operational equivalence. It serves as an opaque search identity, while the structural sketch in §6 is a model-facing feature.

9 Implications for prover design

Separate interfaces. Pretty printing serves human communication and policy conditioning. Deduplication requires an identity with stronger transition-consistency properties. The results support separate representations for these functions: a stable textual observation for the policy and a structurally refined key for the search graph. Version tagging and collision checks remain necessary for a production identity scheme.

Structural conditioning. The paired-ranking result motivates a compact structural channel alongside the rendering. The sketch is member-local, generalizes across unseen theorems within observed mechanisms, and is substantially smaller than shallow `pp.all`. The trained acceptance model and frozen-model intervention together locate the present bottleneck: the signal is learnable under task-specific training, while an unadapted prover gives near-chance action scores. Direct extraction from Lean expressions could remove the prototype’s explicit-printing stage. The digest remains reserved for opaque search identity.

Candidate generation. Member-specific scoring can affect proof search when the candidate set contains a corresponding choice. Increasing

miniCTX beam width raises shared-action coverage but produces no member-specific oracle headroom through $k = 16$. Structurally conditioned generation or targeted proposal mechanisms are therefore the next experimental step. Candidate exposure and candidate-set regret should accompany top- k accuracy when evaluating such systems.

10 Scope of claims

Representation and mechanism. The central claim concerns action sufficiency, for which one verified counterexample is decisive. The 70-class corpus characterizes the phenomenon under the reported search protocol and remains concentrated: 56 classes involve coercion construction, with four `Fin` eta, four local-unfolding, two instance-path, two residual, one `Ne-versus-Not`, and one proof-term class. Causal identification applies to the dominant family. Theorem-grouped prediction measures interpolation within observed mechanisms; smaller and hash-sensitive mechanism holdouts provide preliminary transfer estimates.

Models and search. The frozen-model test evaluates one prompt and scores confirmed tactics. The checkpoint may have encountered Mathlib during training. Candidate-set analyses measure exposure within fixed decoded lists, while proof-rate effects also depend on generation, search policy, and budget. The stress benchmark estimates conditional completeness loss after an admitted alias encounter with an available member-separating continuation. Its 96 pairs derive from one mechanism crossed with logical contexts. Blind sampling and broad paired searches estimate encounter frequency under the evaluated distributions; the single natural loss concerns the aliased node’s bounded continuation.

11 Conclusion

Hidden elaborated structure produces contradictory deterministic labels for identically rendered state-tactic inputs, disproving action sufficiency for the standard textual observation. A compact structural channel makes the missing signal learnable, and controlled retention experiments show that rendering identity can discard an available proof while a structural key preserves it at modest cost. Human-readable rendering and machine-state identity serve distinct design objectives. Treating them as interchangeable fails transition safety; the demonstrated failure is removed by structurally refined identity.

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A Corpus-level census

Across 168,074 auditable Mathlib states, 271 observation-collision classes diverge operationally, but approximately 99% differ only in declaration context: the members occupy different files or positions, so their environments differ. State-weighted prevalence on elaborated-goal identity is 0.03%. An intervention control adding valid declarations while holding the elaborated goal and tactic fixed produced one success-to-failure transition in 497 trials. Cross-file classes cannot be merged by a transposition table, since two theorems never co-occur in one search tree, which is why the main text measures within-search aliasing.

B Human proof trajectories

Across 60,665 traced Mathlib proofs and 168,074 states, only 10 proofs revisit a byte-identical rendering, yielding 12 position pairs. In all 12 the author applies a different tactic at the two positions, yielding zero natural same-tactic pairs for

probing. Linear proof scripts thus rarely create the repeated observation that search produces through reconvergence.

C Fingerprint ladder

One state is read at five levels of printer explicitness by `run_tac` probes: the default rendering φ_{state} ; φ_{shallow} (`pp.all`, notation disabled); φ_{deep} (`pp.deepTerms`); φ_{proofs} ; and φ_{full} . Two implementation properties matter for reproduction: `pp.all` leaves `pp.deepTerms` and proof-term suppression separately configurable, and module identity must be excluded from executable digests or every cross-file pair separates trivially.

D Experimental specification

Policies: `leandojo-lean4-tacgen-byt5-small` and `ntp-mathlib-st-deepseek-coder-1.3b`, state-only input, inputs truncated at 2,300 tokens, deterministic beam search (8 beams), $k=8$; representative sensitivity additionally uses `BFS-Prover-V2-7B` (prompt `{state}:::`, deterministic beam-8, 24 expansions, 300s per member, verification at 72 expansions and 900s). Expansion pool: `simp`, `constructor`, `intro h`, `omega`, plus per-hypothesis `rw [h]` and `simp only [h]` (≤ 16 per state), depth 2. Probe battery: `rfl`, `assumption`, `omega`, `simp`, `norm_num`, `constructor`, `exact?`, `trivial`, `decide`, `simp_all`, `tauto`, `aesop`, plus goal-derived rewrites; the merge-pair battery adds `contrapose!`, `push_neg`, `norm_cast`. Per-probe watchdogs 30–120s; session timeouts 300–900s; three repetitions per member; fixed seeds; shards deduplicated by (file, theorem, tactic index) and confirmed classes by (theorem, rendering hash).

For the direct model-facing test, the frozen `BFS-Prover-V2` checkpoint scores each confirmed tactic by mean conditional token log probability. The structural prompt appends the member-local shallow sketch in a Lean comment before the model-card separator. The ordinary and structural arms use identical tactic tokens. The paired metric assigns one, one half, or zero according to whether the accepting member scores above, ties, or scores below the rejecting member. The theorem-cluster bootstrap uses 10,000 resamples.

The width sweep uses `miniCTX` snapshot `7c00b1175a82e`, one deterministic 16-beam de-

code per class, and a 16-token cap. Prefixes of the single ranked list define each k , ensuring nested candidate sets. All candidates are executed three times per member; a failure includes parser, elaboration, and no-progress errors.

E Visible escape

Escape depth is the first step at which member renderings or terminal outcomes differ under a shared continuation, generated by the policy from the shared observation and applied to every member in lockstep to depth 8, stopping at the first divergence, terminal disagreement, or revisited rendering. Of 59 classes, 19 escape and 18 pass repeat controls. Successor diffs are identical in form: one member’s goal becomes $\forall b, h < b \rightarrow \dots$ while the other’s becomes $\neg \exists b, \dots \wedge b \leq h$.

F Structural digest

One 322-byte `run_tac` probe: `mkForallFVars (lctx.getFVars)` type closes the goal over its local context, converting free variables to de Bruijn indices; `instantiateMVars` resolves assignments; `abstractMVars` canonically renumbers remaining expression and universe metavariables; the resulting term’s 64-bit `Expr.hash` is returned. Validated with three repetitions per member on all 59 classes (59/59 separated, all stable) and as a search key on 39 theorems. The hash approximates normalized structural equality: collisions are possible in principle and the encoding is Lean-version specific.

G Version persistence

The current-stack blind audit samples 60 of 8,302 compiled files uniformly with a fixed seed and traces all 60. They contain 989 theorems and 2,507 tactic states. Root sampling selects 600 states; 570 complete restoration and expansion, while 30 resource or restoration failures remain recorded and are excluded from the incidence denominator. The 98 observed alias classes yield two repeat-confirmed classes on one root, both in `ModuleCat.disjoint_span_sum`. Their two- and four-member histories differ through `rw [huv]` and `simp only [huv]`; shallow `pp.all` and the structural digest distinguish both classes. Seven probes give stable opposite acceptance labels and no probe is nondeterministic.

Eight representative classes were replayed in place on Mathlib master 3b3cddb under Lean v4.33.0-rc1, splicing each proof back into its own file so namespace and variable context are preserved. One reproduces fully, with both histories elaborating, all four members’ renderings byte-identical, and the probe’s successors still diverging. Of the remaining seven, one is a pure rename: the Table 1 witness persists as `lintegral_ofReal_le_lintegral_enorm` with its prefix referencing a renamed lemma, and six are blocked by tactic-level API drift. Every class that remained constructible retained the divergence.

H Node identity across package versions

In the evaluated stack (LeanDojo 2.0.2) `TacticState` is a frozen dataclass whose pretty-printed field participates in equality and hashing while the executable handle is excluded with `compare=False`, and the shipped best-first prover indexes nodes by that value. Directly constructed states with one rendering and different handles compare equal, hash equal, and collide in a dictionary.

We separately audit the published LeanDojo-v2 1.0.9 wheel (SHA-256 635e8f99...) and its active Pantograph backend at commit f8aee320. The active `BaseProver` keeps a history stack and contains no state-indexed transposition dictionary. Pantograph `GoalState` includes `state_id`; two constructed states with equal rendered goals and different identifiers render equally and compare unequally. They are also unhashable because their goal fields are lists. Thus the active v2 prover preserves these distinct live handles and avoids the evaluated rendering-keyed merge.

The v2 wheel also contains a legacy search-tree module whose `InternalNode` compares a `TacticState`, but the wheel omits the interaction-state implementation imported by that module. We therefore base the current-system conclusion on the active Pantograph path. That path supplies the model with `str(GoalState)`, which omits `state_id`; equal renderings can consequently remain observational aliases for the policy. The search-loss claim applies to the evaluated LeanDojo prover, while the action-sufficiency claim applies independently to both textual interfaces.

Identity	nodes	peak queue	time	probe share
rendering	230	135	1.00×	0%
shallow pp.all	233	138	2.79×	64%
structural digest	232	136	1.08×	15%

Table 8: Node-identity cost over 24 rotated-order theorem searches. Time is total wall clock relative to the rendering baseline. All arms prove the same 4 theorems; node and queue differences occur on 2 of 24.

I Node-identity cost

Twenty-four theorems at 12 expansions, arm order rotated with the theorem index, and a baseline arm that omits the probe, since the paired-search harness computes φ_{shallow} in every arm for instrumentation and its recorded wall clock therefore prevents isolation of the key’s cost. Against the baseline’s 230 distinct nodes and peak queue 135, φ_{shallow} retains 233 and 138 and the digest 232 and 136, each differing on 2 of 24 theorems. Expansions are equal (150, 150, 151) because the budget caps them; the counts are therefore uninformative about node overhead. The printing probe costs 0.47 s per node at the median and consumes 64% of its arm’s search time (paired median 2.55×, maximum 12.1×); the digest costs 0.090 s and consumes 15% (paired median 1.27×). Generation and tactic execution are unchanged across arms and all three arms prove the same 4 theorems.

J Retention protocol and reproducibility

Stress construction and admission. Eight acceptance-divergent interventions on Mathlib 3b3cddb/Lean v4.33.0-rc1 are crossed with 12 proposition contexts, including conjunction, disjunction, equality with `True`, double negation, and duplicated goals. Each context has a logical introduction sequence that exposes the base pair’s acceptance-sensitive probe as the proof-producing step. Lean compilation then enforces three admission conditions: byte-identical default renderings, distinct structural digests, and closure of the abbreviated member with rejection on the unfolded member. All 96 candidates pass. The harness compiles 192 capture declarations and a matrix of 17,472 state-action labels. Capture declarations log the state before an explicit `sorry`; no state-action closure label relies on `sorry`.

Retention and proposal protocol. The comparison begins after both branches reach the admitted alias. Rendering identity retains the first arrival,

while rendering plus digest retains both. The targeted generator emits the pair’s known continuation followed by `rfl`. The context-bank generator emits the eight continuations lifted through the same proposition context, followed by `rfl`. The generic-first generator prefixes `rfl`, `simp`, and `norm_num` to that bank. Budgets are 2, 4, 8, and 12 actions. Both orders are enumerated, and 200 uniformly randomized orders per pair and configuration use seed 731991. The primary interval resamples pairs; a second interval clusters the 12 contexts by their eight base templates. Exact order enumeration eliminates Monte Carlo error. The bootstrap intervals measure sensitivity to the constructed pair set.

Supplementary controls. The earlier eight-pair protocol gives five order-dependent losses with the pair-specific grammar and the same five with one fixed union grammar over 194 compilations. For the natural case, injecting both histories from their common root and continuing with BFS-Prover-V2 proves every retention arm through another intermediate within 2–4 expansions. The observed natural loss is therefore confined to the aliased node’s continuation.

Step-prover configuration. Checkpoint ByteDance-Seed/BFS-Prover-V2-7B, snapshot 533d59fc6b5e, with its own tokenizer; prompt is the state’s rendering followed by `:::` per the model card; float16 on Apple MPS. Decoding is deterministic beam search, 8 beams, 8 return sequences, at most 64 new tokens, early stopping; the continuation is decoded without special tokens, text after any re-emitted `:::` is kept, and candidates are deduplicated preserving the best sequence score. Tactic execution carries a 60 s watchdog; a watchdog interrupt abandons the class as channel desynchronization. The search queue orders by negated cumulative sequence score with insertion order breaking ties. Per-class first candidate lists, per-search reached-state hashes, and proof traces are in the released run files, together with the three repetition runs and the triple-budget run for the sensitive class.

K Auxiliary candidate-selection analyses

Fixed-candidate reranking. For each of the 146 restored members, we execute the BFS-Prover-V2 top-8 list three times, yielding 3,504 compiler calls and 1,168 state-candidate labels. The lists are iden-

Ordering	top-1	top-3	MRR	first rank
7B generator	.658	.856	.763	1.575
text-only reranker	.863	.877	.878	1.209
+ structural sketch	.877	.877	.886	1.149
transferred sketch	.644	.863	.761	1.500

Table 9: Out-of-fold selection on 146 alias members. “First rank” is the mean first accepted rank among the 134 members covered by the fixed top-8.

Policy	$ C $	covered	ambiguous	rank 1	headroom > 0
ReProver	59	55	2	0	0
miniCTX	55	41	2	1	0
BFS-7B	59	56	1	0	0

Table 10: Candidate exposure. “Covered” counts classes with an accepted candidate for every member; “ambiguous” counts classes containing an acceptance-divergent candidate.

tical across members of every alias class, and all repetitions agree. Logistic rerankers receive the tactic, default state, original rank, and optionally the member-local sketch. Five folds group by theorem, and all reported selection metrics use out-of-fold scores.

The structural increment over text-only reranking is $+0.0084$ MRR and $+1.4$ top-1 percentage points in the primary configuration. Its theorem-bootstrap 95% interval is $.0000$ – $.0257$. The increment is $+0.0023$, -0.0002 , and $+0.0108$ under the ten-fold, 16,384-dimensional, and 2,048-dimensional alternatives. A separate classifier trained only on the 184 contradiction examples gives $.7607$ MRR when transferred to these candidate lists and ranks the sole acceptance-divergent 7B candidate in the wrong member direction.

Candidate-set headroom. For candidate set A and class C , the member-aware oracle averages $\max_{a \in A} r(s_i, a)$ over members. The shared-action oracle maximizes the mean acceptance of one action across members. Their difference measures member-specific headroom within A and has no monotonic relationship with the full tactic language.

For BFS-Prover-V2, the member-aware and shared-action oracles both cover 134/146 members. The corresponding equality holds in every fully covered class for all three generators. These fixed candidate sets provide almost no opportunity for a member-aware score to alter proof availability.

L Validity

Three controls address hazards that produced false positives during development. After any watchdog interrupt the session is discarded, because channel desynchronization can return an earlier probe’s answer. Every divergence must be self-consistent over three repetitions per member in a fresh session; this removed 11 of 13 candidates supported only by nondeterministic `aesop` (Limperg and From, 2023). Finally, escape is reported relative to the tactic battery, and rendering-based canonicalization was compared with a separating fingerprint on 369 paired roots. The artifact retains exclusions, repeat tables, and resource-failure verdicts.

M Artifact

The supplement contains all harnesses and per-class outcomes, including the representation, direct frozen-model scoring, and auxiliary reranking rows, folds, transfer test, and robustness runs; intervention files; policy candidates; search traces; pinned versions; and seeds. One command reproduces each table from released records, with live modes for the motivating witness and current intervention.